ORANGE Report

SAMPLE
Demo WGS

PRIMARY TUMOR	PURITY	PLOIDY	FIT METHOD	QC
Skin	99% (97%-100%)	3.00 (2.96-3.05)	NORMAL	PASS

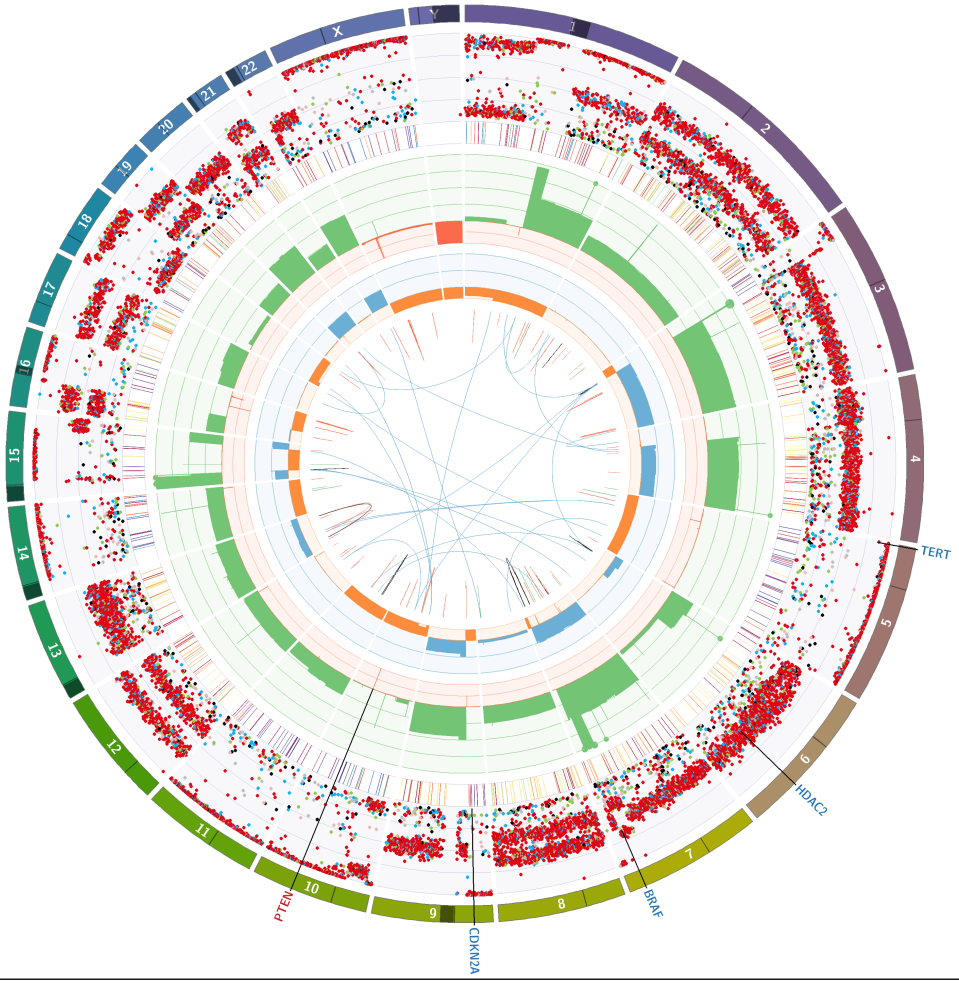
PIPELINE VERSION	GENOME VERSION	SEQUENCING TYPE	PIPELINE	SAMPLES	DATE ANALYSED
OA_V3_0	V38	ILLUMINA	WHOLE GENOME	TUMOR / NORMAL	2026-06-16

Driver Summary

Somatic variant:	7 (BRAF, CDKN2A, HDAC2, TERT)
Somatic copy number:	1 (PTEN)
Somatic disruption:	2
Germline variant:	None
Germline copy number:	None
Germline disruption:	None
Fusion drivers:	None
Viral presence:	None
Whole genome duplicated:	Yes
DPYD status:	*1 HOM (Normal Function)

Genome Wide Biomarkers

Microsatellite indels per Mb:	0.1 (Stable)
Tumor mutations per Mb:	14.4 (High)
Tumor mutational load:	198 (High)
HR deficiency score:	0.0 (Proficient)
LOH proportion:	15%
Number of SVs:	125
CUPPA cancer type:	Skin: Melanoma (100%)



Somatic Findings

Small Variants (8)

GENE	POSITION	HGVS	AF	DEPTH	COPIES	HOTSPOT	BIALL	CLONAL	DRIVER
BRAF	chr7:140753336	c.1799T>A [p.Val600Glu]	0.62	237	3.8 of 6.1	Yes	2%	100%	100%
CDKN2A	chr9:21971154	c.203_204delCG [p.Ala68fs]	1.00	96	2.0 of 2.0	Near	100%	100%	100%
CDKN2A	chr9:21971154	c.246_247delCG [p.Gly83fs]	1.00	96	2.0 of 2.0	Near	100%	100%	100%
TERT	chr5:1295113	c.-125_-124delCCinsTT	0.86	73	1.7 of 2.0	Yes	93%	100%	100%
HDAC2	chr6:113943504	c.1225C>T [p.Arg409*]	0.45	94	1.0 of 2.2	No	2%	100%	33%
GRIN2A	chr16:10180333	c.79G>T [p.Ala27Ser]	0.07	121	0.2 of 3.1	No	2%	0%	14%
SF3B1	chr2:197402055	c.2153C>T [p.Pro718Leu]	0.65	115	2.0 of 3.0	No	2%	100%	14%
TP63	chr3:189886541	c.1497G>T [p.Met499Ile]	0.55	150	2.2 of 4.0	No	2%	100%	0%

Amplifications and Deletions (1)

LOCATION	GENE	TYPE	RANGE	MIN CN	MAX CN	REL CN	ARM CN	DRIVER
chr10q23.31	PTEN	DEL	PARTIAL	0.0	2.0	0.0	2.0	HIGH

Fusions (0)

NONE

Disruptions (1)

GENE	POSITION	ZYGOSITY	CONTEXT	TYPE	JCN	UNDISRUPTED CN	DRIVER
PTEN	chr10:87940542 - chr10:87952584	HOM	Intron 5 - Intron 6	DEL	2.0	0.0	LOW

Viruses (0)

NONE

Arm Copy Number Abberations

CHROMOSOME	ARM	TYPE	CN	REL CN	DRIVER
chr1	Q	GAIN	3.9	1.4	HIGH
chrY	P	LOSS	0.0	0.0	HIGH
chrY	Q	LOSS	0.0	0.0	HIGH
chr1	P	LOSS	2.3	0.7	LOW
chr3	Q	GAIN	4.0	1.3	LOW
chr4	P	LOSS	2.0	0.7	LOW
chr4	Q	GAIN	3.8	1.3	LOW
chr5	P	LOSS	2.0	0.7	LOW
chr5	Q	LOSS	2.0	0.7	LOW
chr7	P	GAIN	4.0	1.3	LOW

THE TABLE CONTINUES ON THE NEXT PAGE

Arm Copy Number Abberations

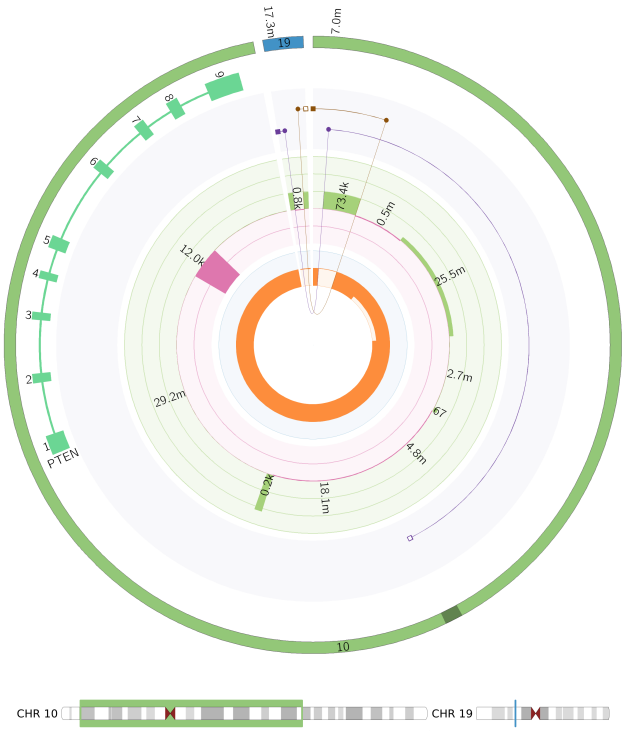
CONTINUED FROM THE PREVIOUS PAGE

CHROMOSOME	ARM	TYPE	CN	REL CN	DRIVER
chr7	Q	GAIN	4.0	1.4	LOW
chr9	Q	GAIN	3.8	1.3	LOW
chr10	P	LOSS	2.2	0.7	LOW
chr10	Q	LOSS	2.0	0.7	LOW
chr16	Q	LOSS	2.0	0.7	LOW
chr18	P	LOSS	2.0	0.7	LOW
chr18	Q	LOSS	2.2	0.7	LOW
chr20	P	GAIN	4.0	1.3	LOW
chr20	Q	GAIN	4.0	1.3	LOW
chr22	Q	GAIN	4.0	1.3	LOW
chrX	P	GAIN	1.9	1.3	LOW
chrX	Q	GAIN	1.9	1.3	LOW

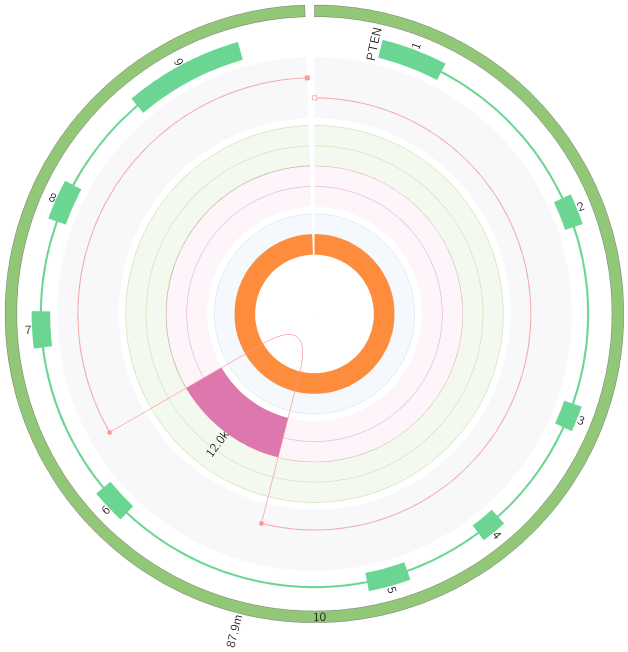
Signature Allocations (12)

SIGNATURE	ETIOLOGY	ALLOCATION	PERCENT
Sig7	Ultraviolet light exposure	23623	60%
Sig2	APOBEC activity	3862	10%
Sig8	Unknown	2312	6%
Sig18	Reactive oxygen species	2309	6%
Sig4	Tobacco smoking	2236	6%
Sig12	Unknown	1644	4%
Sig11	Temozolomide treatment	1529	4%
Sig17	Unknown	1012	3%
Sig28	Unknown	562	1%
MISALLOC	-	4984	13%

Structural Driver Plots (2)


17.3m
7.0m
19
9
8
7
6
5
4
3
2
1
PTEN
10
0.8k
73.4k
0.5m
25.5m
2.7m
67
4.8m
18.1m
0.2k
29.2m
12.0k

CHR 10
CHR 19


PTEN
1
2
3
4
5
6
7
8
9
10
87.9m
12.0k


CHR 10

Germline Findings

Small Variants (0)

NONE

Amplifications and Deletions (0)

NONE

Disruptions (0)

NONE

Chromosomal aberrations (0)

NONE

Pharmacogenetics (1)

GENE	HAPLOTYPE	GENOTYPE	FUNCTION	LINKED DRUGS	SOURCE
DPYD	*1	HOM	Normal Function	5-Fluorouracil;Capecitabine;Tegafur	PHARMGKB

Immunology

HLA Class I Alleles

ALLELE	QC STATUS	REF FRAGS	TUMOR FRAGS	TUMOR CN	SOMATIC MUTATIONS
A*01:01	PASS	258	706	2.0	NONE
A*01:01	PASS	258	707	1.4	NONE
B*08:01	PASS	216	640	1.4	NONE
B*40:02	PASS	212	751	2.0	NONE
C*03:04	PASS	249	803	2.0	NONE
C*07:01	PASS	225	621	1.4	NONE

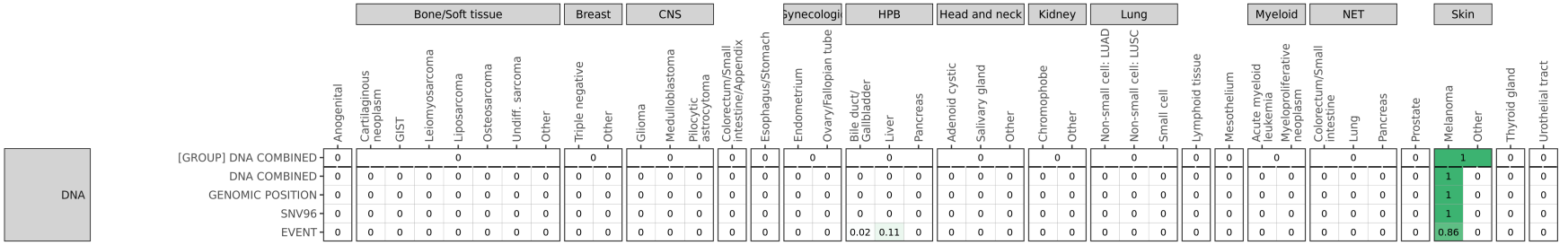
HLA Class II Alleles

ALLELE	QC STATUS	REF FRAGS	TUMOR FRAGS	TUMOR CN	SOMATIC MUTATIONS
DPA1*01:03	PASS	127	440	2.0	NONE
DPA1*02:01	PASS	103	296	1.4	NONE
DPB1*01:01	PASS	150	374	1.4	NONE
DPB1*02:01	PASS	129	491	2.0	NONE
DQA1*05:01	PASS	92	293	2.0	NONE
DQA1*05:05	PASS	99	298	1.4	NONE
DQB1*02:01	PASS	106	314	1.4	NONE
DQB1*03:01	PASS	113	415	2.0	NONE

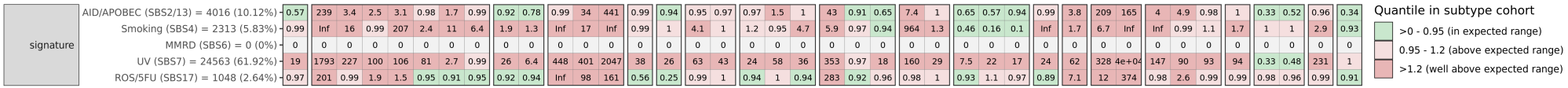
Tissue of Origin

Probabilities by classifier

Cancer group (strips) and subtype (label)



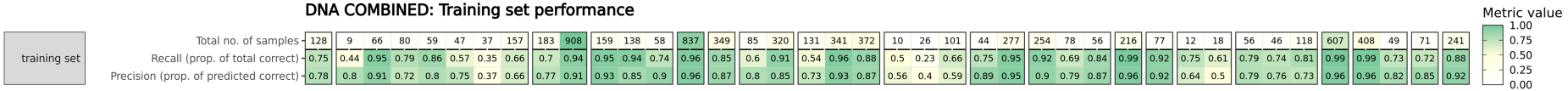
SNV96: Mutational signatures



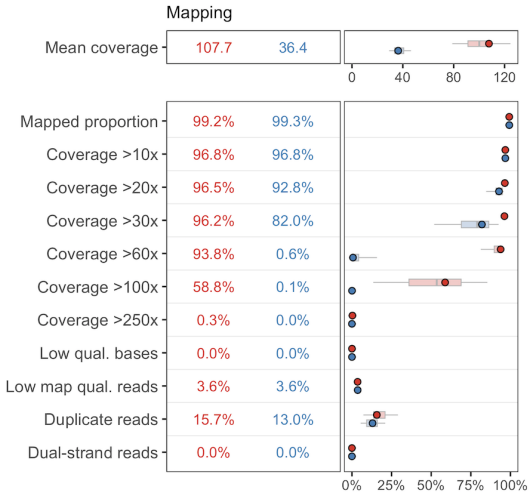
EVENT: Feature contributions

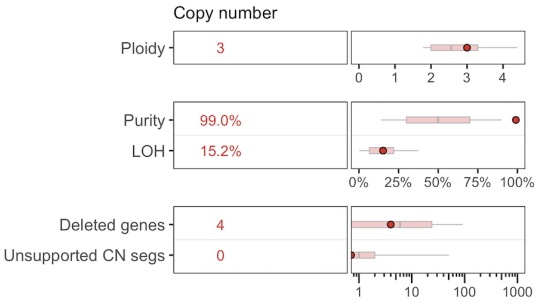


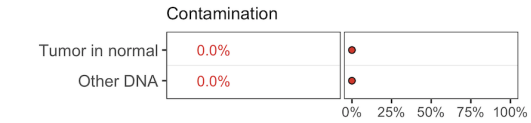
DNA COMBINED: Training set performance

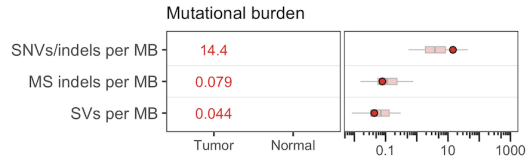


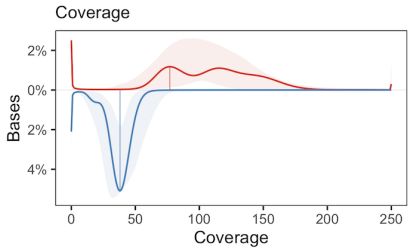
Quality Control

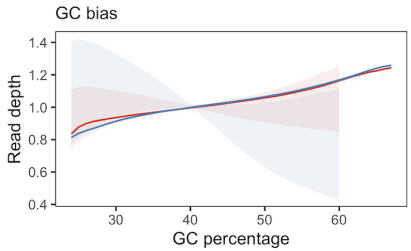


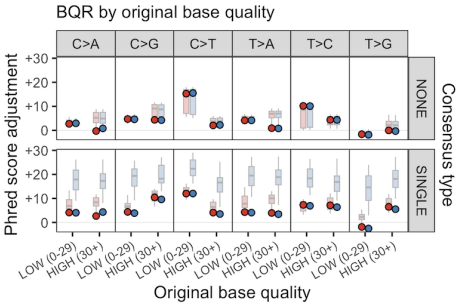


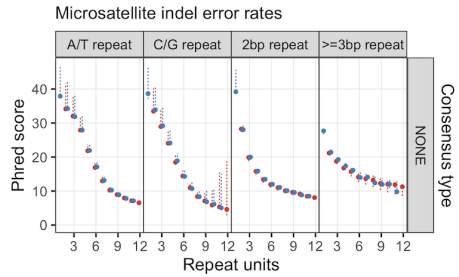


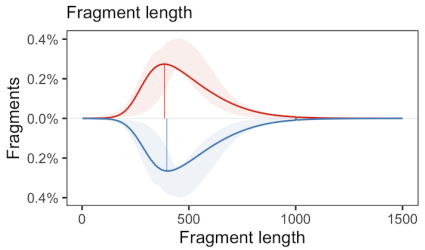


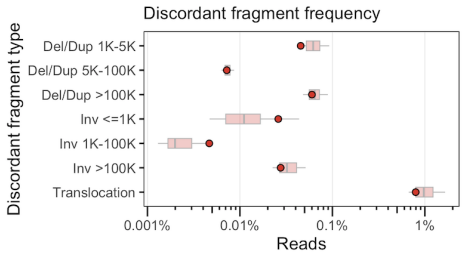


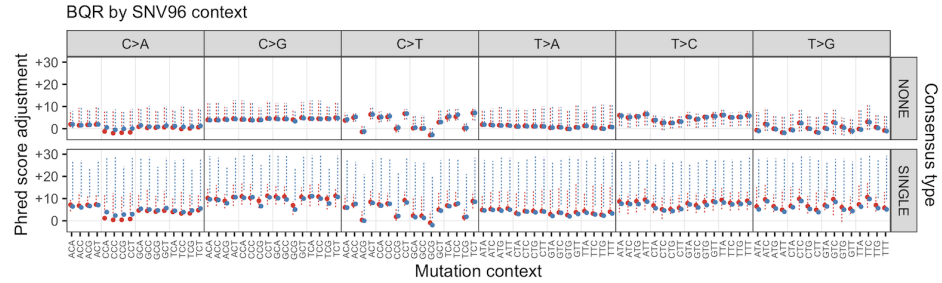


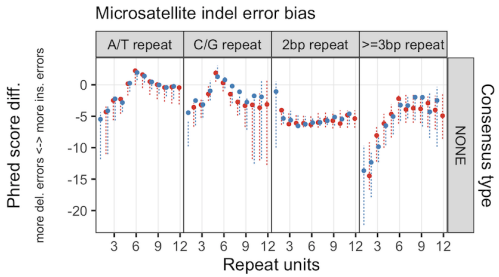


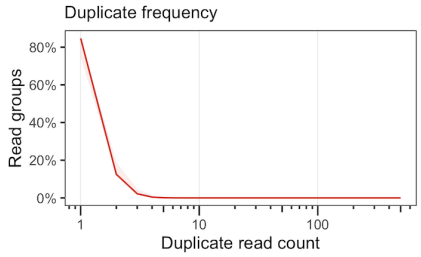


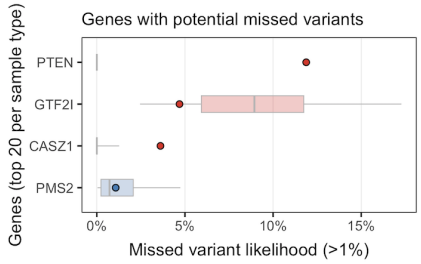


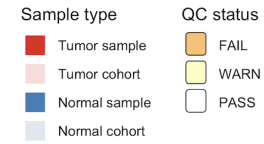












Purity and Ploidy

